

Statistical Walkthrough

Figures and Code Explained with Basic + Intermediate Statistical Terminology

Companion to the Tissue-Specific DEG Analysis | Generated by Biomni (Phylo) | 2026-07-09

This document walks through each figure and key code block from the tissue-specific DEG analysis, explaining the statistical concepts in basic and intermediate terms. Each section covers one figure or code block, describing what it shows, what statistical test or concept underlies it, how to read the results, and what the numbers mean in plain language.

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Part I: Figures Explained

Each figure below is reproduced with a full statistical explanation. We cover what the plot displays, what statistical concept it illustrates, how to interpret the visual elements, and what conclusions we can draw.

Figure 1 — Stacked Bar Chart: DEG Counts by Organ

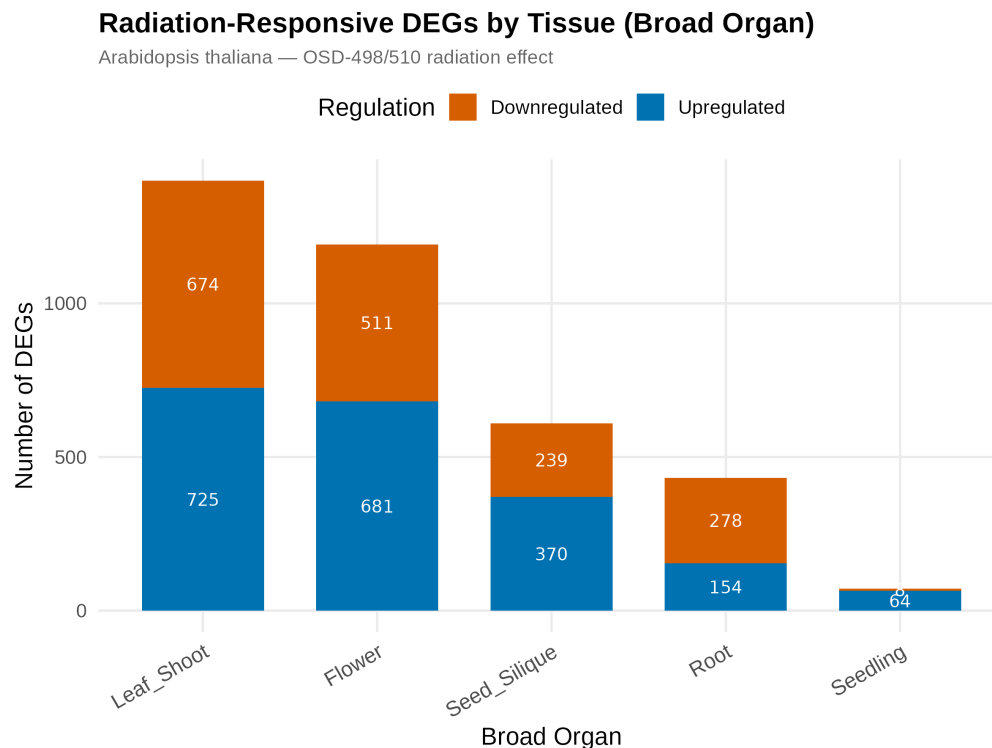


Figure 1: Number of upregulated (blue) and downregulated (orange) DEGs per broad organ.

What This Graph Shows

A stacked bar chart displays the count of differentially expressed genes (DEGs) for each of the five broad organ categories. Each bar is split into two colors: blue for genes that increased in expression after radiation (upregulated) and orange for genes that decreased (downregulated). The total height of each bar represents the total number of tissue-specific DEGs assigned to that organ.

Statistical Concept: Frequency Distribution

This is a **frequency distribution** — a count of how many observations (genes) fall into each category (organ). In statistics, a frequency distribution is the most basic way to summarize categorical data. The bars show the **absolute frequency** (raw count) for each category. The blue/orange split shows a **conditional frequency** — the count of up vs. down within each organ.

BASIC TERM: What is a frequency?

A frequency is simply how many times something occurs. If 1,399 genes in the Leaf_Shoot category are DEGs, then the frequency of Leaf_Shoot DEGs is 1,399. A frequency distribution shows these counts for all categories at once, making it easy to compare groups visually.

How to Read It

- **Bar height** = total DEGs for that organ. Taller bars mean more radiation-responsive genes were assigned to that tissue.

- **Blue vs. orange proportion** = direction of response. If blue dominates, that tissue's response is primarily activation (upregulation). If orange dominates, the response is primarily suppression (downregulation).
- **Numbers inside bars** = exact counts for each segment.

What the Data Tell Us

Leaf_Shoot has the tallest bar (1,399 DEGs), meaning more radiation-responsive genes are normally active in leaf and shoot tissues than anywhere else. The blue and orange segments are roughly equal in Leaf_Shoot, indicating a balanced mix of up- and downregulation. In contrast, Root shows more orange than blue (278 down vs. 154 up), suggesting radiation predominantly suppresses root gene expression. Seedling is almost entirely blue (64 up vs. 8 down), indicating a strong activation response.

INTERMEDIATE TERM: Why counts alone can be misleading

Raw counts don't account for the fact that some organs have more tissue-specific genes to begin with. Leaf_Shoot has 3,288 tissue-specific genes in the atlas; Root has only 1,195. So Leaf_Shoot having more DEGs could simply be because it has more genes total. To determine whether the *rate* of differential expression differs (not just the count), we need the statistical tests in Figure 5. This is the difference between **absolute frequency** (raw counts) and **relative frequency** (proportions or rates).

Figure 2 — Volcano Plots: Magnitude vs. Significance

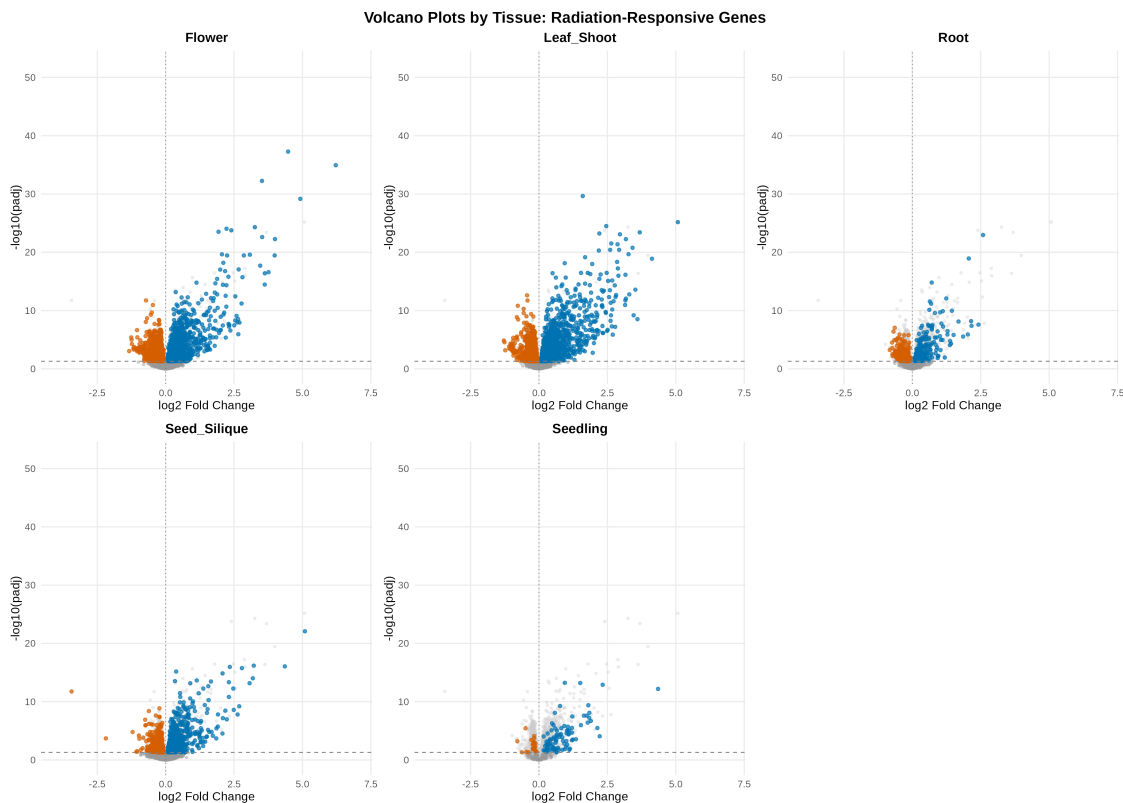


Figure 2: Five volcano plots (one per organ). Blue = upregulated DEGs, orange = downregulated, grey = non-DEGs or constitutive genes.

What This Graph Shows

A volcano plot is a scatter plot where each point is one gene. The x-axis shows the **log2 fold change** (magnitude and direction of expression change), and the y-axis shows the **-log10 adjusted p-value** (statistical significance). Five panels show the same plot separately for each broad organ, with constitutive genes (broadly expressed) shown as grey background points for context.

Statistical Concept: Effect Size vs. Significance

Volcano plots display two fundamental statistical quantities simultaneously:

- **Effect size** (x-axis): The log2 fold change measures how much the gene's expression changed. A value of +2 means the expression quadrupled; -1 means it halved. This is the *magnitude* of the biological effect.
- **Statistical significance** (y-axis): The -log10(padj) measures how confident we are that the change is real (not due to chance). Higher values = more confident. The dashed horizontal line marks padj = 0.05, the conventional significance threshold.

BASIC TERM: What is a p-value?

A p-value answers the question: 'If there were truly no difference between irradiated and control plants, how likely is it that we would observe a difference this large or larger just by random chance?' A small p-value (typically < 0.05) means the observed difference is unlikely to be due to chance, so we conclude the difference is real. The adjusted p-value (padj) corrects for the fact that we are testing thousands of genes at once.

INTERMEDIATE TERM: Why $-\log_{10}(\text{padj})$ instead of padj directly?

P-values for significant genes can be extremely small (e.g., 5.34×10^{-38}). On a linear scale, all these tiny values would cluster near zero and be indistinguishable. Taking $-\log_{10}$ transforms them: $-\log_{10}(0.05) = 1.3$, $-\log_{10}(0.001) = 3$, $-\log_{10}(10^{-38}) = 38$. Now the most significant genes are at the top of the plot, spread out and visible. This is called a **logarithmic transformation**, commonly used to spread out values that span many orders of magnitude.

How to Read It

- **Points above the dashed line** = statistically significant ($\text{padj} < 0.05$).
- **Points to the right of $x=0$** = upregulated (expression increased). Points to the left = downregulated (expression decreased).
- **Points in the top-right corner** = genes that are both highly significant AND strongly upregulated — the most biologically interesting genes.
- **Points in the top-left corner** = highly significant AND strongly downregulated.

What the Data Tell Us

Each organ panel shows a different pattern. Leaf_Shoot and Flower have many points spread across both sides of $x=0$, indicating both up- and downregulation. Root has more points on the left (downregulation), consistent with the bar chart. Seed_Silique shows a cluster of blue points on the right (upregulation). The vertical spread (significance) is similar across panels, but the horizontal spread (effect size) and direction differ — this is the tissue specificity we are quantifying.

Figure 3 — Heatmap: Expression Specificity of Top DEGs

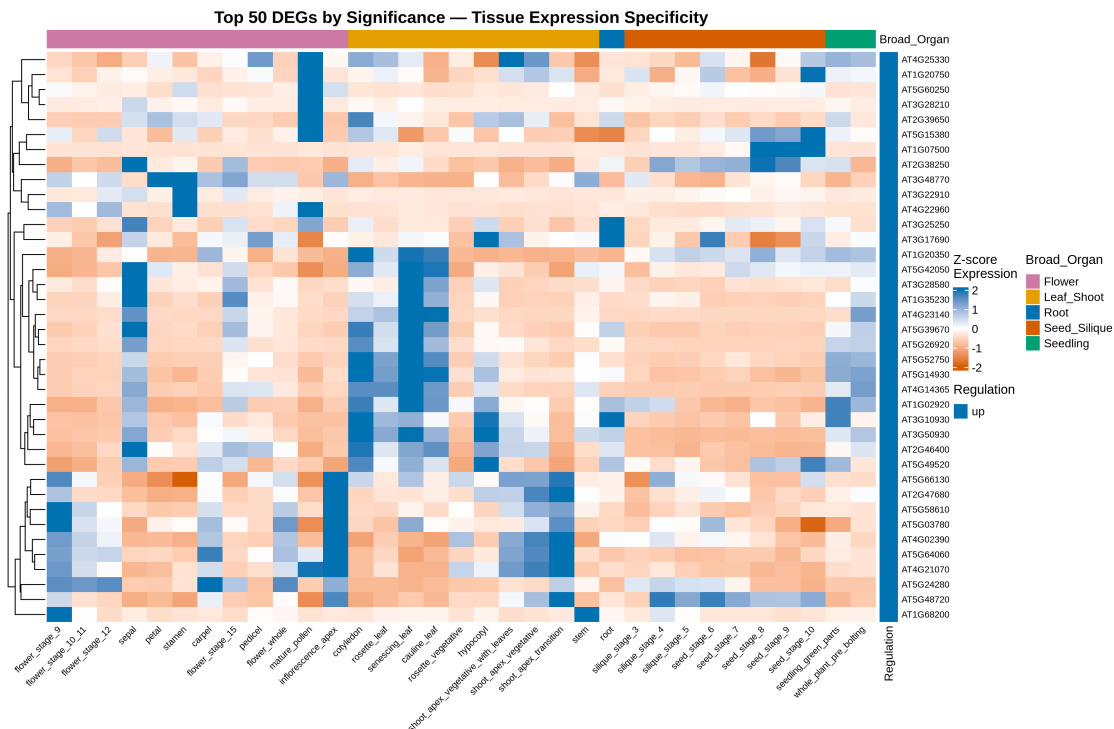


Figure 3: Z-scored expression of the 50 most significant DEGs across 33 sub-tissues. Top bar = broad organ; right bar = regulation direction.

What This Graph Shows

A heatmap displays expression levels as colors. Each row is one of the 50 most significant DEGs (sorted by padj); each column is one of 33 sub-tissues. The color intensity shows how highly that gene is expressed in that tissue, relative to its own average across all tissues. The top annotation bar groups columns by broad organ; the right annotation bar shows whether each gene was up- or downregulated in response to radiation.

Statistical Concept: Z-Score Normalization

The colors represent **z-scores**, not raw expression values. A z-score measures how many standard deviations a value is above or below the gene's mean expression across all tissues.

$$z = (\text{expression_in_tissue} - \text{mean_expression}) / \text{standard_deviation}$$

$z = 0 \rightarrow$ expression is exactly at the gene's average

$z = +2 \rightarrow$ expression is 2 standard deviations above average

$z = -2 \rightarrow$ expression is 2 standard deviations below average

BASIC TERM: Why use z-scores instead of raw values?

Different genes have very different baseline expression levels. Gene A might be expressed at 1,000 units everywhere; gene B at 10 units everywhere. If we plotted raw values, gene A would always be darker than gene B, making it impossible to see tissue-specific patterns. Z-scoring rescales each gene so its average is 0 and its variation is comparable to other genes. Now we can see patterns: a red block means 'this gene is expressed much higher here than its own average,' regardless of its absolute level.

INTERMEDIATE TERM: Standard deviation as a spread measure

The standard deviation (SD) measures how spread out a gene's expression is across tissues. If a gene is expressed at similar levels everywhere, its SD is small, and even a modest difference will produce a large z-score. If a gene varies wildly across tissues, its SD is large, and only big differences produce notable z-scores. This means z-scores account for each gene's natural variability — a form of **standardization** that makes genes comparable.

How to Read It

- **Red blocks** = the gene is expressed above its personal average in that tissue.
- **Blue blocks** = the gene is expressed below its personal average.
- **Rows that are red in only one organ block** = tissue-specific genes (high Tau).
- **Rows that are uniformly colored** = broadly expressed genes (low Tau).
- **The right-side annotation** lets you check whether upregulated vs. downregulated DEGs tend to come from specific tissues.

What the Data Tell Us

The heatmap reveals that the most significant radiation-responsive genes come from diverse tissue backgrounds. Some rows show sharp red blocks in only one organ (e.g., a row that is red only in the Root columns), confirming these are tissue-specific genes. Others show patchy patterns across multiple organs, suggesting broader expression. The clustering of rows (dendrogram on the left) groups genes with similar tissue expression profiles, revealing that upregulated and downregulated DEGs are interspersed — radiation affects genes from many tissue contexts.

Figure 4 — UpSet Diagram: Cross-Organ DEG Overlaps

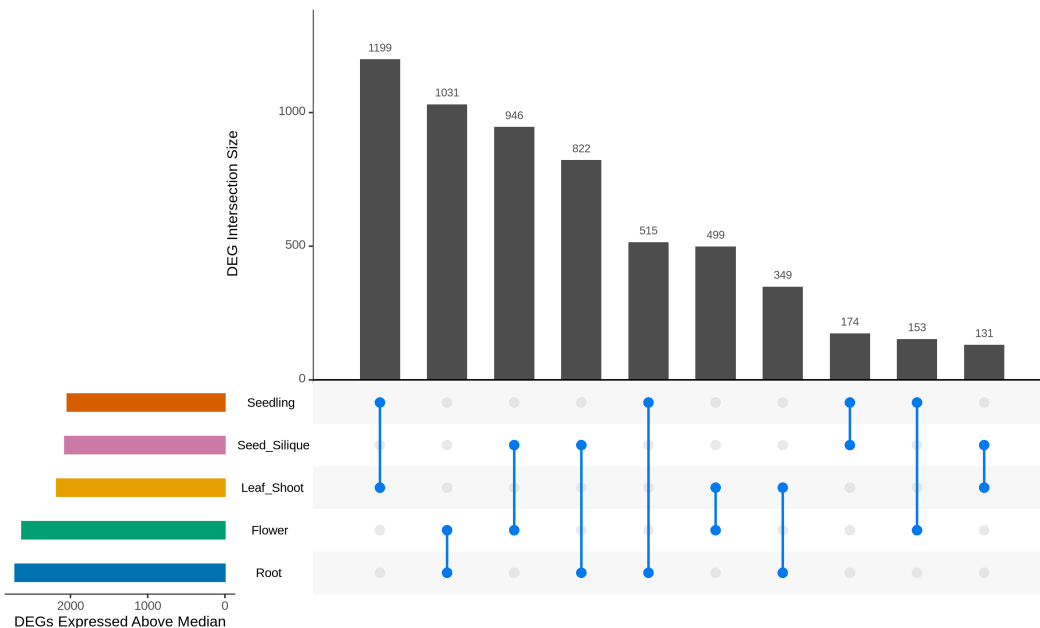


Figure 4: UpSet diagram showing how many DEGs are expressed above median in each combination of organs.

What This Graph Shows

An UpSet diagram is an alternative to a Venn diagram for showing **set intersections**. Each DEG can be 'expressed above its own median' in multiple organs. The left bar chart shows the total number of DEGs expressed above median in each organ. The main bar chart shows how many DEGs are shared across each combination of organs. The matrix of connected dots below indicates which organs are included in each intersection.

Statistical Concept: Set Intersection Analysis

A **set** is a collection of items (here, DEGs expressed in a given organ). The **intersection** of two sets is the collection of items that belong to both. For example, the intersection of 'DEGs expressed in Root' and 'DEGs expressed in Flower' is the set of DEGs that are active in both organs. UpSet diagrams scale better than Venn diagrams when there are more than 3-4 sets.

BASIC TERM: What is a median?

The median is the middle value when you sort a set of numbers. If a gene's expression across 5 organs is [10, 20, 30, 40, 50], the median is 30. We use 'above median' as a threshold: an organ counts as 'expressing' the gene if its expression value is above the gene's median. This is a **non-parametric** threshold — it doesn't assume any particular distribution, just that values above the midpoint are 'relatively high' for that gene.

INTERMEDIATE TERM: Why not a Venn diagram?

Venn diagrams work for 2-3 sets but become unreadable with 5 sets (the circles overlap in ways that make regions too small to label). The UpSet diagram solves this by using a matrix of dots to encode set membership: a filled dot means 'included in this set,' and connected dots define an intersection. The bar height shows the size of that intersection. This is a **scalable** visualization for set intersection analysis.

How to Read It

- **Left bars** = total DEGs expressed above median in each organ (set sizes).
- **Main bars** = number of DEGs in each specific intersection (combination of organs).
- **Dot matrix** = which organs are included in each intersection. Filled dots connected by a line = the organs whose overlap is being counted in that bar.

What the Data Tell Us

The largest intersections typically involve 2-3 organs, meaning many DEGs are co-expressed across multiple tissues. This is expected: genes involved in core stress responses (like DNA repair) are active in many tissues. The single-organ bars (only one dot filled) represent DEGs that are expressed above median in only one organ — these are the most tissue-restricted. The distribution of intersection sizes tells us whether radiation-responsive genes tend to be organ-specific or broadly active.

Figure 5 — Forest Plot: Tissue Enrichment Statistics

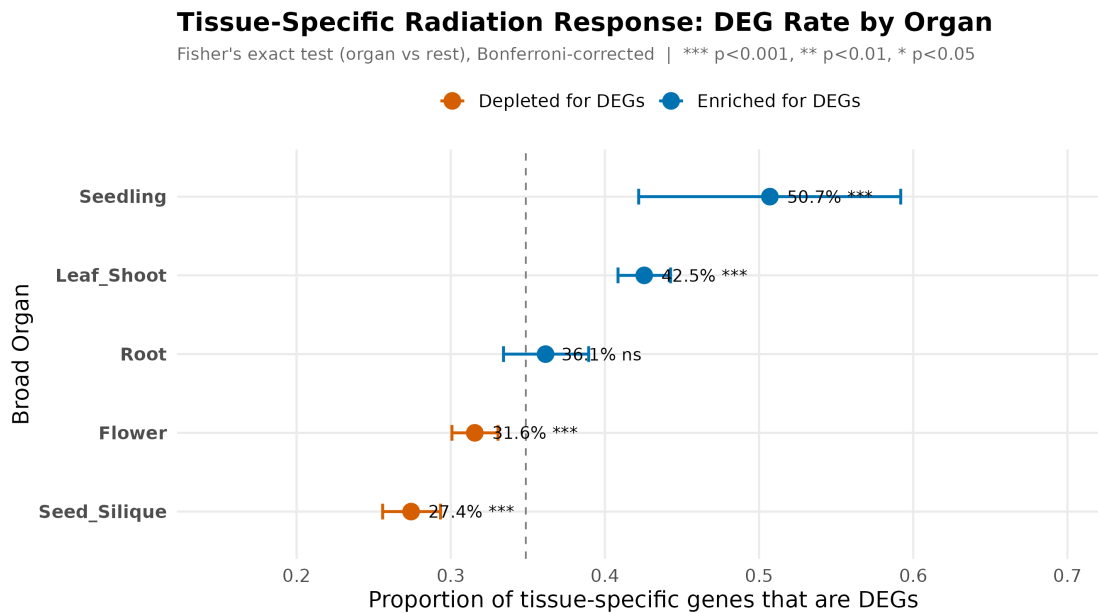


Figure 5: DEG rate per organ with 95% confidence intervals. Dashed line = overall rate. Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

What This Graph Shows

A forest plot displays the **proportion** of tissue-specific genes that became DEGs for each organ, along with **95% confidence intervals** (horizontal lines) and significance markers. The dashed vertical line shows the overall DEG rate across all organs (the rate expected if tissue had no effect). Points to the right of the dashed line have higher-than-expected DEG rates (enriched); points to the left have lower-than-expected rates (depleted).

Statistical Concept: Proportion, Confidence Interval, and Hypothesis Testing

This figure brings together three core statistical concepts:

- **Proportion** (the dots): The DEG rate is a proportion — the fraction of tissue-specific genes that are DEGs. For example, Seedling's rate of 50.7% means 72 out of 142 seedling-specific genes are DEGs. Proportions range from 0 to 1 (or 0% to 100%).
- **95% confidence interval** (the horizontal lines): The CI tells us the range of values within which we are 95% confident the true DEG rate lies. Narrow lines = precise estimate (large sample); wide lines = uncertain estimate (small sample). Seedling has the widest CI because it has the fewest tissue-specific genes (142).
- **Hypothesis testing** (the significance stars): Each organ is tested against the null hypothesis that its DEG rate equals the overall rate. *** means $p < 0.001$ (very strong evidence against the null), ** means $p < 0.01$, * means $p < 0.05$, and 'ns' means not significant.

BASIC TERM: What is a confidence interval?

When we calculate that 42.5% of Leaf_Shoot genes are DEGs, that's our best estimate from the sample we have. But the true rate in the entire population of Leaf_Shoot genes might be slightly different. A 95% confidence interval says: 'If we repeated this experiment many times, 95% of the time the true rate would fall within this range.' For Leaf_Shoot, the CI is 40.9%-44.3%, meaning we're fairly confident the true rate is close to our estimate.

INTERMEDIATE TERM: What is the odds ratio?

The odds ratio (OR) compares the odds of being a DEG in one group vs. another. $OR = 1$ means no difference; $OR > 1$ means the first group has higher odds (enriched); $OR < 1$ means lower odds (depleted). For Leaf_Shoot, $OR = 1.62$ means a tissue-specific Leaf_Shoot gene has 1.62 times the odds of being a DEG compared to a gene from any other organ. For Seed_Silique, $OR = 0.65$ means its genes have 35% lower odds of being DEGs. The OR is the test statistic used in Fisher's exact test (see Code Block 3).

How to Read It

- **Dots** = estimated DEG rate for each organ.
- **Horizontal lines** = 95% confidence interval. If the line does not cross the dashed vertical line, the organ's rate is significantly different from the overall rate.
- **Color** = blue = enriched (more DEGs than expected), orange = depleted (fewer).
- **Stars** = statistical significance after Bonferroni correction for multiple testing.

What the Data Tell Us

Seedling (50.7%) and Leaf_Shoot (42.5%) are significantly enriched for DEGs — their confidence intervals are entirely to the right of the dashed line. Seed_Silique (27.4%) and Flower (31.6%) are significantly depleted — their CIs are entirely to the left. Root (36.2%) is not significantly different from the overall rate (34.9%), as its CI overlaps the dashed line. This means leaf/shoot and seedling tissues show a stronger transcriptional response to radiation than expected, while reproductive tissues (seed/silique) and flowers show a weaker response.

Part II: Key Code Blocks Explained

This section explains the statistical reasoning behind the five most important code blocks in the analysis. For each block, we show the code, explain what statistical test or concept it implements, and walk through the output.

Code Block 1 — The Tau Tissue-Specificity Index

The Code

```
compute_tau <- function(expr_matrix) {
  expr_matrix[expr_matrix < 0] <- 0
  apply(expr_matrix, 1, function(x) {
    mx <- max(x, na.rm = TRUE)
    if (mx == 0) return(NA)
    n <- sum(!is.na(x))
    sum(1 - (x / mx), na.rm = TRUE) / (n - 1)
  })
}

tau_subtissue <- compute_tau(sub_tissue_expr)
specificity <- ifelse(tau_subtissue >= 0.6, "tissue_specific", "constitutive")
```

What This Code Does

This function computes the **Tau index** for every gene, measuring how tissue-specific its expression pattern is. The `apply()` function with `MARGIN = 1` iterates over each row (gene) of the expression matrix. For each gene, it normalizes all tissue expression values by dividing by the maximum, subtracts each from 1, sums these, and divides by $(n - 1)$ where n is the number of tissues.

Statistical Concept: Tissue-Specificity Metric

The Tau index is a **specificity metric** — a single number that summarizes how unevenly a gene's expression is distributed across tissues. It is based on the idea of normalizing each tissue's expression by the maximum expression observed for that gene:

```
Tau = sum(1 - x_i / max(x)) / (n - 1)

x_i = expression of gene in tissue i
max(x) = highest expression across all tissues
n = number of tissues

Range: 0 (ubiquitous) to 1 (perfectly tissue-specific)
```

BASIC TERM: What does 'normalizing by the max' mean?

Imagine a gene expressed at levels [100, 10, 5, 2, 1] across 5 tissues. The max is 100. Dividing each by 100 gives [1.0, 0.1, 0.05, 0.02, 0.01]. Now subtract each from 1: [0, 0.9, 0.95, 0.98, 0.99]. Sum = 3.82. Divide by $(5-1) = 4$. Tau = 0.955. This gene is almost exclusively expressed in tissue 1 — it's highly tissue-specific. Now imagine a gene expressed at [100, 98, 95, 92, 90]. Same max (100), but the ratios are [1.0, 0.98, 0.95, 0.92, 0.90]. Subtract from 1: [0, 0.02, 0.05, 0.08, 0.10]. Sum = 0.25. Divide by 4. Tau = 0.063. This gene is expressed nearly everywhere — it's constitutive.

INTERMEDIATE TERM: Why Tau instead of other specificity metrics?

Several metrics exist for tissue-specificity: Tau, the Gini coefficient, the entropy-based specificity score (H), and the tissue specificity index (TSI). Tau is preferred because it is (1) robust to noise — small fluctuations in low-expression tissues have minimal impact, (2) bounded between 0 and 1, making interpretation straightforward, and (3) does not require any distributional assumptions about the expression data. The threshold of 0.6 is a widely used convention in the plant genomics literature, balancing sensitivity (catching real tissue-specific genes) with specificity (not misclassifying broadly expressed genes).

The Output

After computing Tau, each gene is classified as either **tissue-specific** (Tau $\geq$ 0.6) or **constitutive** (Tau < 0.6). Genes not present in the AtGenExpress atlas are labeled **unannotated**. For this dataset:

Classification Results

Tissue-specific DEGs: 3,704 (53.4% of all DEGs)

Constitutive DEGs: 2,115 (30.5%)

Unannotated DEGs: 1,123 (16.2%)

Mean Tau (all atlas genes): 0.662

Median Tau: 0.661

Code Block 2 — Chi-Square Test of Independence

The Code

```
# Build contingency table: organ x DEG status
contingency <- table(tissue_specific$predominant_broad_organ,
tissue_specific$deg_flag)

# Chi-square test of independence
chi_result <- chisq.test(contingency)
print(chi_result)
```

What This Code Does

First, `table()` builds a **contingency table** — a matrix that counts how many genes fall into each combination of organ (rows) and DEG status (columns: yes/no). Then `chisq.test()` performs a **chi-square test of independence**, which asks: 'Is DEG status independent of tissue assignment, or does the probability of being a DEG depend on which tissue a gene belongs to?'

Statistical Concept: Chi-Square Test of Independence

The chi-square test compares the **observed** counts in each cell of the contingency table to the **expected** counts we would see if tissue and DEG status were truly independent (the null hypothesis). If the observed counts deviate substantially from expected, we reject the null hypothesis and conclude that tissue matters.

```
Chi-square statistic:  $X^2 = \sum ((\text{observed} - \text{expected})^2 / \text{expected})$ 

observed = actual count in each cell
expected = count expected if tissue and DEG status were independent
           = (row total x column total) / grand total

Degrees of freedom:  $df = (\text{rows} - 1) \times (\text{cols} - 1) = (5 - 1) \times (2 - 1) = 4$ 
```

BASIC TERM: What is a null hypothesis?

The null hypothesis (H0) is the 'nothing is going on' assumption. Here, H0 says: 'The probability of a gene being a DEG is the same regardless of which tissue it belongs to.' The test calculates how likely the observed data would be if this boring scenario were true. If the data would be very unlikely under H0 (small p-value), we reject H0 and conclude that tissue does matter. If the data are plausible under H0 (large p-value), we fail to reject H0 — we don't have evidence that tissue matters.

INTERMEDIATE TERM: What are residuals?

The chi-square test also produces **residuals** — the standardized difference between observed and expected counts for each cell: $(\text{observed} - \text{expected}) / \sqrt{\text{expected}}$. A positive residual means more DEGs than expected (enriched); negative means fewer (depleted). Residuals above +2 or below -2 are conventionally considered noteworthy. For this data, Leaf_Shoot has a residual of +7.5 (strongly enriched) and Seed_Silique has -5.9 (strongly depleted), immediately revealing which tissues drive the overall significant result.

The Output

Chi-Square Test Results

X-squared = 174.2

Degrees of freedom = 4

p-value = 1.31×10^{-36}

Conclusion: REJECT H0 ($p \ll 0.05$)

Tissue assignment and DEG status are NOT independent.

There IS a statistically significant tissue-specific radiation response.

Key residuals (DEG column):

Leaf_Shoot: +7.5 (strongly enriched)

Seed_Silique: -5.9 (strongly depleted)

Flower: -3.4 (depleted)

Seedling: +3.2 (enriched)

Root: +0.7 (near expected)

Code Block 3 — Fisher's Exact Test (Per-Tissue)

The Code

```
for (org in organs) {
  # 2x2 table: this organ vs all others x DEG vs non-DEG
  org_yes <- contingency[org, "yes"]
  org_no <- contingency[org, "no"]
  rest_yes <- sum(contingency[, "yes"]) - org_yes
  rest_no <- sum(contingency[, "no"]) - org_no

  mat <- matrix(c(org_yes, org_no, rest_yes, rest_no), nrow = 2)
  ft <- fisher.test(mat, alternative = "two.sided")
}

# Bonferroni correction for 5 tests
fisher_results$padj <- p.adjust(fisher_results$p_value, method = "bonferroni")
```

What This Code Does

For each organ, the code builds a 2x2 table comparing that organ to all other organs combined, crossed with DEG status (yes/no). It then runs `fisher.test()` to test whether that organ's DEG rate differs significantly from the rest. Finally, `p.adjust()` applies the **Bonferroni correction** for the fact that we are running 5 tests (one per organ).

Statistical Concept: Fisher's Exact Test

Fisher's exact test is an alternative to the chi-square test for 2x2 tables. While the chi-square test approximates the p-value using a theoretical distribution, Fisher's test calculates the **exact** p-value by enumerating all possible tables that could arise under the null hypothesis. This makes it more accurate for small sample sizes, where the chi-square approximation may be unreliable.

BASIC TERM: What is the odds ratio?

The odds ratio (OR) is the test statistic from Fisher's test. 'Odds' is different from probability: if 72 out of 142 seedling genes are DEGs, the probability is $72/142 = 0.507$, but the odds are $72/(142-72) = 72/70 = 1.03$ (ratio of 'yes' to 'no'). The odds ratio compares the odds in two groups: $OR = \text{odds}(\text{seedling}) / \text{odds}(\text{rest})$. $OR = 1$ means no difference; $OR > 1$ means the first group has higher odds; $OR < 1$ means lower odds. For Seedling, $OR = 1.94$, meaning seedling-specific genes have nearly double the odds of being DEGs.

INTERMEDIATE TERM: Why Bonferroni correction?

When we run multiple statistical tests, the probability of getting at least one false positive increases. If we run 5 tests at $\alpha = 0.05$, the chance of at least one false positive is $1 - (0.95)^5 = 23\%$. The **Bonferroni correction** controls this by dividing the significance threshold by the number of tests: $0.05 / 5 = 0.01$. A result is significant only if its adjusted p-value (padj) is below 0.05. This is the simplest and most conservative multiple testing correction. Alternatives include Benjamini-Hochberg (FDR control), which is less conservative and more commonly used in genomics, but Bonferroni is appropriate here because we have only 5 tests.

The Output

Organ	DEG Rate	vs Rest	Odds Ratio	padj	Direction
Seedling	50.7%	34.7%	1.94	6.0e-04	Enriched ***
Leaf_Shoot	42.5%	31.4%	1.62	1.4e-27	Enriched ***
Root	36.2%	34.7%	1.07	1.00	ns
Flower	31.6%	36.7%	0.80	5.2e-07	Depleted ***
Seed_Silique	27.4%	36.8%	0.65	2.7e-16	Depleted ***

Root is the only organ that is not significantly different from the rest (padj = 1.00). Its DEG rate (36.2%) is close to the overall rate (34.9%), so the data are consistent with the null hypothesis for root specifically — even though the overall chi-square test (Code Block 2) showed that tissue matters in general.

Code Block 4 — Pairwise Fisher's Exact Tests

The Code

```
for (i in 1:(length(organs) - 1)) {
  for (j in (i + 1):length(organs)) {
    mat <- matrix(c(
      contingency[organs[i], "yes"], contingency[organs[i], "no"],
      contingency[organs[j], "yes"], contingency[organs[j], "no"]
    ), nrow = 2)
    ft <- fisher.test(mat, alternative = "two.sided")
  }
}

# Bonferroni correction for C(5,2) = 10 pairwise tests
pairwise_results$padj <- p.adjust(pairwise_results$p_value, method =
  "bonferroni")
```

What This Code Does

This nested loop compares every pair of organs directly. With 5 organs, there are $C(5,2) = 10$ possible pairs. For each pair, it builds a 2x2 table (organ1 x organ2, crossed with DEG status) and runs Fisher's exact test. The Bonferroni correction now divides by 10 instead of 5, because we are running 10 tests.

Statistical Concept: Pairwise Comparisons and Multiple Testing

The per-tissue test (Code Block 3) compares each organ to 'all others combined.' But the user's question — 'are roots responding more than shoots?' — requires a **direct pairwise comparison**. This code block answers that by testing each organ against each other organ individually.

BASIC TERM: What is a combination $C(n,2)$?

If you have 5 organs and want to compare every pair, the number of unique pairs is '5 choose 2' = $C(5,2) = 5! / (2! \times 3!) = 10$. The nested loop generates these 10 pairs: (Flower, Leaf_Shoot), (Flower, Root), (Flower, Seed_Silique), (Flower, Seedling), (Leaf_Shoot, Root), and so on. Each pair gets its own Fisher's exact test.

INTERMEDIATE TERM: The multiple testing problem escalates

With 10 pairwise tests at $\alpha = 0.05$, the probability of at least one false positive under the global null is $1 - (0.95)^{10} = 40\%$. The Bonferroni correction becomes more stringent: $0.05 / 10 = 0.005$. Despite this stricter threshold, 9 of 10 comparisons remain significant, indicating robust differences between organs. The only non-significant pair is Seedling vs. Leaf_Shoot ($\text{padj} = 0.57$), likely because both are enriched and their confidence intervals overlap.

The Output

The pairwise results establish a clear ranking of radiation responsiveness:

Pairwise Ranking (DEG rate, highest to lowest)

1. Seedling: 50.7%
2. Leaf_Shoot: 42.5%
3. Root: 36.2%
4. Flower: 31.6%
5. Seed_Silique: 27.4%

9 of 10 pairwise comparisons significant ($p_{adj} < 0.05$)

Only non-significant: Seedling vs. Leaf_Shoot ($p_{adj} = 0.57$)

Direct answer to 'are roots responding more than shoots?':

Leaf_Shoot > Root: OR = 1.31, $p_{adj} = 0.001$

Yes — leaf/shoot genes are significantly more likely to be DEGs.

Code Block 5 — Direction-of-Response Chi-Square Test

The Code

```
# For tissue-specific DEGs only, test if up/down ratio
# differs across organs
direction_table <- table(ts_degs$predominant_broad_organ,
ts_degs$regulation)

# Chi-square test: is up/down ratio independent of tissue?
chi_dir <- chisq.test(direction_table)
```

What This Code Does

This builds a different contingency table: organ (rows) x regulation direction (columns: up/down), using only the DEGs (not all genes). The chi-square test asks whether the **proportion of up vs. down regulation** is the same across all organs, or whether some tissues are predominantly upregulated while others are predominantly downregulated.

Statistical Concept: Testing for Qualitative Differences

The previous tests asked 'how many genes respond?' (quantity). This test asks 'in which direction do they respond?' (quality). Even if two organs have the same number of DEGs, they could differ in whether those DEGs are mostly up or mostly down. This is a fundamentally different biological question: upregulation suggests activation of pathways, while downregulation suggests suppression.

BASIC TERM: What does 'up' vs. 'down' mean biologically?

When a gene is 'upregulated' after radiation, it means the plant is producing more of that gene's protein — typically because the pathway is being activated (e.g., a DNA repair gene is upregulated because the cell needs more repair protein). 'Downregulated' means less protein is being made — the pathway is being suppressed (e.g., a cell cycle gene is downregulated because the cell pauses division to repair damage first). The balance of up vs. down tells us whether a tissue is primarily 'turning things on' or 'shutting things down' in response to radiation.

INTERMEDIATE TERM: Why a separate test for direction?

The first chi-square test (Code Block 2) used a 5x2 table (organ x DEG yes/no). It could detect that organs differ in DEG rate, but it treated all DEGs the same regardless of direction. This second test uses a 5x2 table (organ x up/down) among DEGs only. It can detect that organs differ in response direction even if they have similar DEG counts. For example, Root and Flower have similar DEG rates (36.2% vs. 31.6%), but Root is 64% downregulated while Flower is 57% upregulated — a qualitative difference that the first test would miss.

The Output

Direction-of-Response Test Results

X-squared = 112.3, df = 4, p-value = 2.33×10^{-23}

Conclusion: REJECT H_0 ($p \ll 0.05$)

The up/down regulation ratio DIFFERS significantly across tissues.

Per-organ up/down ratios:

Seedling: 88.9% up (predominantly UPREGULATED)

Seed_Silique: 60.8% up

Flower: 57.1% up

Leaf_Shoot: 51.8% up (roughly balanced)

Root: 35.6% up (predominantly DOWNREGULATED)

Biological interpretation:

Root: radiation suppresses gene expression (cell cycle arrest)

Seedling: radiation activates gene expression (stress response)

Part III: Statistical Glossary

Quick reference for the statistical terms used throughout this document, organized from basic to intermediate.

Term	Level	Definition
Frequency	Basic	How many times something occurs. A frequency distribution shows counts for all categories.
Proportion	Basic	A fraction or percentage: the count in one category divided by the total count.
Median	Basic	The middle value when data are sorted. Half the values are above it, half below.
Null hypothesis (H0)	Basic	The 'nothing is going on' assumption. The test asks: could the data plausibly arise if H0 were true?
P-value	Basic	The probability of observing data this extreme (or more) if H0 were true. Small p-value = evidence against H0.
Statistical significance	Basic	When $p < 0.05$ (conventional threshold), we reject H0 and call the result 'significant.'
Confidence interval (CI)	Basic	A range of values that likely contains the true parameter. A 95% CI means 95% of repeated experiments would capture the true value.
Odds ratio (OR)	Intermediate	Ratio of odds between two groups. OR = 1: no difference; OR > 1: group 1 has higher odds; OR < 1: lower odds.
Effect size	Intermediate	The magnitude of a difference (e.g., log2 fold change). Distinct from significance, which measures confidence.
Contingency table	Intermediate	A matrix counting observations across two categorical variables (e.g., organ x DEG status).
Chi-square test	Intermediate	Tests whether two categorical variables are independent by comparing observed vs. expected counts.
Fisher's exact test	Intermediate	An exact test for 2x2 tables; calculates the precise p-value rather than approximating it. Preferred for small samples.
Degrees of freedom (df)	Intermediate	The number of independent pieces of information. For a contingency table: $(rows-1) \times (cols-1)$.
Residuals	Intermediate	Standardized difference between observed and expected counts. $ residual > 2$ indicates that cell deviates from expectation.
Bonferroni correction	Intermediate	Divides the significance threshold by the number of tests to control the family-wise error rate. Conservative but simple.
Z-score	Intermediate	Number of standard deviations a value is from the mean. Used to standardize variables on different scales.
Logarithmic transformation	Intermediate	Applying log (e.g., log10 or log2) to spread out values spanning many orders of magnitude. Used in volcano plots.

Term	Level	Definition
Tau index	Intermediate	A specificity metric from 0 (ubiquitous) to 1 (tissue-specific). Threshold 0.6 separates tissue-specific from constitutive genes.
Multiple testing problem	Intermediate	Running many tests increases false positives. Corrections (Bonferroni, BH-FDR) adjust p-values to control this.

Key Takeaways from the Statistical Analysis

1. **Tissue matters** (chi-square $p = 1.3 \times 10^{-36}$): the probability of a gene being a DEG depends significantly on which tissue it belongs to.
2. **Leaf/Shoot and Seedling are enriched** for DEGs (higher response rate than expected); **Flower and Seed/Silique are depleted** (lower rate). Root is not significantly different from the overall rate.
3. **9 of 10 pairwise comparisons are significant**, establishing a clear ranking: Seedling > Leaf_Shoot > Root > Flower > Seed_Silique.
4. **Direction differs by tissue** (chi-square $p = 2.3 \times 10^{-23}$): roots are predominantly downregulated (suppression), while seedlings are predominantly upregulated (activation). This is a qualitative difference, not just a quantitative one.